RAHUL D R

Electronics & Communication Engineering Student
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PROFILE SUMMARY

Electronics and Communication Engineering student with hands-on experience in PCB assembly, hand soldering, circuit continuity testing, and embedded systems. Practical exposure to schematic reading and board-level troubleshooting through an internship at an electronics manufacturing company. Currently building foundational knowledge of PCB design workflows and actively developing skills in industry-standard EDA tools (Mentor Xpedition, Altium, Cadence Allegro). Quick learner, detail-oriented, and eager to grow into a PCB Design Engineer role through structured, hands-on training.

TECHNICAL SKILLS

Hardware & Assembly: Hand Soldering, PCB Component Assembly, Breadboarding, Component-Level Troubleshooting, Precision Wiring, Relay Logic

Testing & Diagnostics: Multimeter Testing, Circuit Continuity Checks, Schematic Reading, Board-Level Fault Tracing

PCB Design (Learning): Building foundational knowledge of schematic entry and PCB layout; actively upskilling in Mentor Xpedition, Altium Designer, and Cadence Allegro

Microcontrollers & Modules: Arduino Uno/Nano, ESP8266 Wi-Fi Module, MFRC522 RFID, Relays, Motors/Actuators

Communication Protocols: UART, SPI, I2C, CAN (Simulated via TS Master)

Programming & Tools: C, Python (Basic), Arduino IDE, MATLAB Simulink, AutoCAD (Basic)

TECHNICAL INTERNSHIPS

Electronics Engineering Intern | *SLM Technologies*

- Assembled components on PCBs, identified electronics hardware, and performed precision hand soldering.
- Used circuit schematics and a multimeter to run continuity checks and trace board-level issues — direct, hands-on exposure to the PCB assembly and verification process.

Networking Infrastructure Intern | *BSNL (Bharat Sanchar Nigam Limited)*

- Studied telecom routing setups — network switches, routing basics, MAC tables, and static IP allocation.
- Shadowed network technicians during maintenance tasks, signal diagnostics, and hardware troubleshooting.

Web UI Intern | *Cadibal Technologies*

- Learned HTML layouts and basic front-end design rules to understand web interfaces.
- Studied how hardware/microcontroller data can be structured for display on user-facing web dashboards.

HARDWARE & ENGINEERING PROJECTS

SAE Drone Development Project | *Design & Fabrication Project*

- Built the physical frame structure and managed electrical wiring and power distribution layout for a prototype UAV.
- Calculated battery capacity requirements and analyzed BLDC motor specifications to maximize flight time.

Smart RFID Attendance System | *Personal Project*

- Built a working attendance tracker by interfacing an MFRC522 RFID reader module with an Arduino board.
- Used SPI and UART protocols to read hardware data packets and log unique card IDs automatically.

Embedded Medicine Reminder Unit | *Academic Project*

- Created a medication alarm unit using Arduino internal timers to trigger accurate schedules and alerts.
- Wired and programmed active buzzers and status LEDs for clear physical notifications on reminders.

EDUCATION

B.E. in Electronics and Communication Engineering — Expected Graduation: 2027

Knowledge Institute of Technology, Salem | Current CGPA: 6.90 / 10 (up to 5th sem)

Higher Secondary Certificate (HSC) — Completed

Holy Cross Matriculation School, Salem | State Board

AREAS OF INTEREST

PCB Design, Embedded C, Automotive Systems, Robotics